

OYYK, Kazakhstan

WMO: 382030

Lat: 43.783N

Lon: 70.933E

Elev: 366

StdP: 97.00

Time Zone: 6.00 (E06)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.9	-18.6	-25.0	0.4	-21.2	-21.6	0.6	-17.9	10.0	-0.5	9.0	-0.3	1.0	N/A	0.465

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.2	37.3	18.9	35.8	18.3	34.4	17.8	20.2	32.9	19.2	32.5	18.5	31.9	2.4	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	12.0	24.7	14.7	10.9	23.5	13.2	9.9	23.4	59.3	32.4	56.1	32.5	53.5	32.0	26.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	7.1	5.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	-5.4	-4.0	4.0	12.1	18.7	24.2	26.9	24.3	18.2	10.1	2.3	-2.1
	DBStd	12.27	6.26	6.61	6.23	4.77	4.45	3.44	3.01	3.22	4.21	4.99	6.11	5.49
	HDD10.0	1769	476	392	198	31	3	0	0	0	1	58	236	374
	HDD18.3	3471	734	624	444	192	48	3	1	2	52	259	480	632
	CDD10.0	2089	0	0	12	95	274	426	523	444	247	61	6	0
	CDD18.3	750	0	0	0	6	61	179	266	187	49	3	0	0
	CDH23.3	9879	0	0	0	95	729	2277	3673	2388	657	60	0	0
	CDH26.7	4935	0	0	0	26	265	1118	2067	1197	252	10	0	0
Wind	WSAvg	1.6	1.3	1.5	1.8	2.2	1.9	1.6	1.4	1.5	1.4	1.3	1.3	1.5
Precipitation	PrecAvg	213	17	20	27	35	24	9	6	2	7	18	24	24
	PrecMax	305	65	66	52	93	62	28	28	28	29	61	55	60
	PrecMin	126	1	3	3	1	7	0	0	0	0	0	1	3
	PrecStd	48	13	15	13	27	15	7	7	5	8	14	14	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.7	13.1	20.8	29.9	34.1	38.5	40.1	37.8	35.2	28.0	20.1	14.3
		MCWB	5.9	8.9	12.2	16.6	17.7	20.1	20.4	18.2	18.0	14.8	11.0	7.9
	2%	DB	6.5	8.7	17.9	26.0	31.8	36.2	37.9	36.1	32.2	24.9	16.2	10.2
		MCWB	3.7	5.1	10.5	14.7	17.0	18.5	19.3	18.0	16.8	13.2	9.4	6.4
	5%	DB	4.0	6.1	15.4	22.8	29.7	34.4	36.4	34.7	29.7	22.2	13.1	7.4
		MCWB	2.1	3.4	9.2	13.1	16.1	17.6	18.8	17.5	15.5	11.8	8.4	4.3
	10%	DB	2.0	3.7	12.9	20.2	27.5	32.6	34.9	32.8	27.3	19.5	10.6	5.0
		MCWB	0.5	1.3	7.7	11.8	15.3	16.9	18.3	16.8	14.4	10.5	6.8	2.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.6	8.5	13.3	17.9	20.2	21.2	23.0	20.1	19.0	15.4	12.1	8.2
		MCDB	9.5	12.5	18.1	27.7	29.9	33.9	35.6	32.4	33.9	26.7	18.0	13.3
	2%	WB	3.8	5.4	11.3	15.9	18.4	19.4	20.7	18.8	17.2	13.6	10.3	6.7
		MCDB	5.9	8.2	16.8	23.5	28.5	33.2	33.7	32.7	30.8	23.6	15.5	10.1
	5%	WB	2.3	3.5	9.5	14.2	17.2	18.4	19.6	18.0	16.1	12.3	8.7	4.8
		MCDB	4.0	6.1	15.0	21.5	26.6	31.9	32.9	32.9	29.1	21.4	12.6	7.1
	10%	WB	0.7	1.5	8.0	12.7	16.1	17.7	18.8	17.3	14.7	10.8	7.0	3.1
		MCDB	1.9	3.6	12.4	19.1	25.6	30.5	32.4	31.1	26.3	18.7	10.5	4.7
Mean Daily Temperature Range	5% DB	MDBR	7.7	8.9	10.0	11.9	13.2	13.8	14.2	14.6	15.1	14.0	9.7	7.7
		MCDBR	8.7	11.5	14.0	16.3	16.4	16.1	16.5	17.0	18.1	17.3	14.2	11.9
		MCWBR	6.6	7.8	7.6	7.7	6.1	5.2	5.4	5.6	7.3	8.7	8.4	8.3
	5% WB	MCDBR	8.2	10.0	13.3	14.0	13.9	14.5	15.4	15.6	17.3	16.6	13.1	10.4
		MCWBR	6.2	7.4	7.7	7.5	6.2	5.4	5.8	5.4	7.6	8.5	8.1	7.4
Clear-Sky Solar Irradiance	taub	0.359	0.395	0.444	0.468	0.455	0.444	0.433	0.398	0.376	0.407	0.401	0.352	
	taud	2.038	1.991	1.974	1.965	2.056	2.125	2.168	2.249	2.267	2.084	2.045	2.072	
	Ebn at Noon	716	761	778	797	821	831	836	854	845	741	659	681	
	Edh at Noon	109	137	159	173	163	153	145	129	117	123	109	96	
All-Sky Solar Radiation	RadAvg	1.63	2.57	3.83	5.35	6.85	7.55	7.43	6.74	5.28	3.30	1.88	1.33	
	RadStd	0.24	0.30	0.29	0.53	0.36	0.31	0.34	0.22	0.22	0.28	0.20	0.14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	+0.80	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page